Monitoring Task

Task Description:

Install Prometheus and Grafana on a Linux EC2 machine, connect Prometheus to Grafana, and create a dashboard to view metrics.

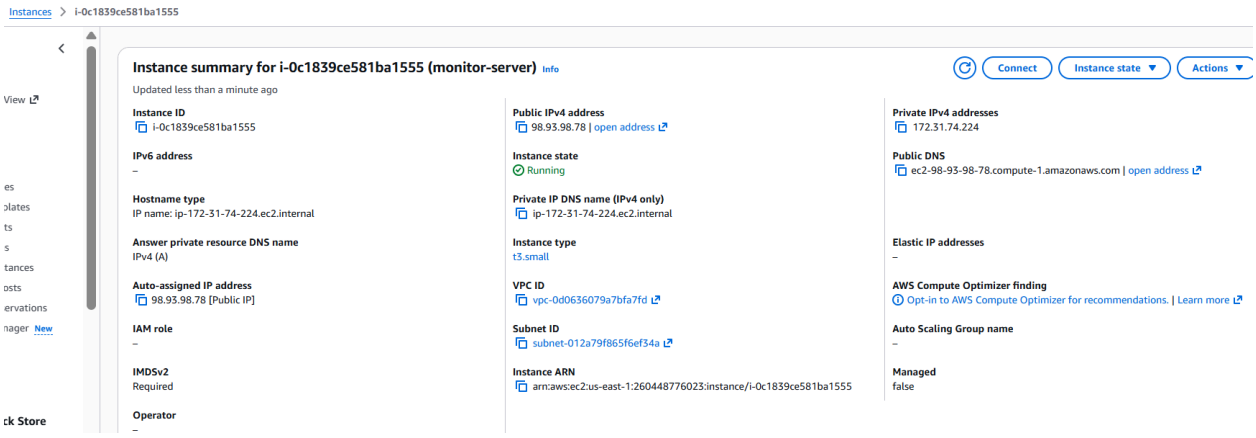
Techstacks needs to be used :

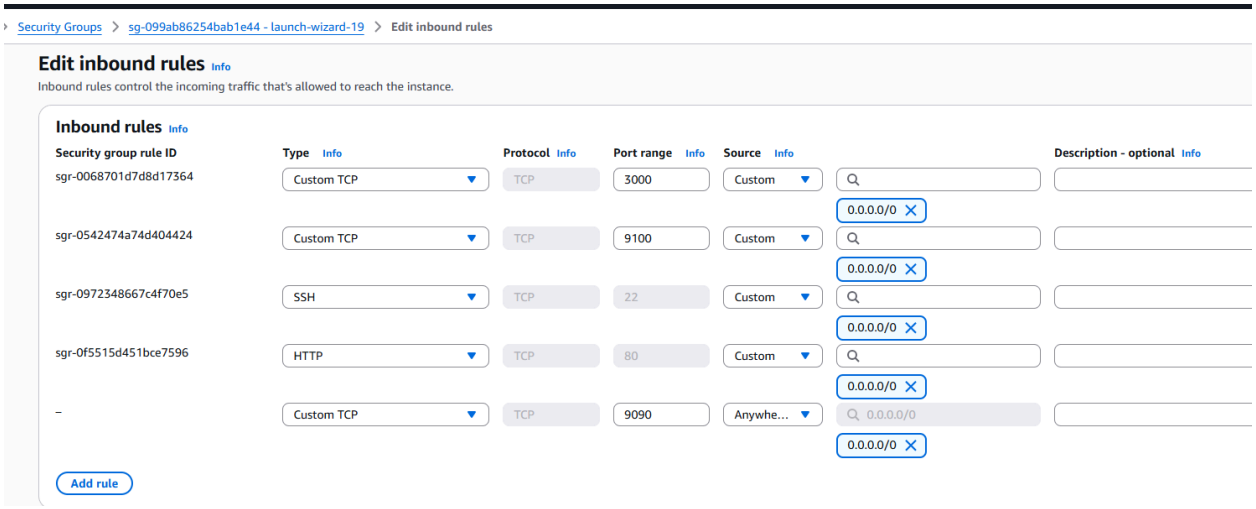
- AWS EC2
- Prometheus
- Grafana
- Node Exporter

Ec2 server created with security inbound rules:

Security Group:

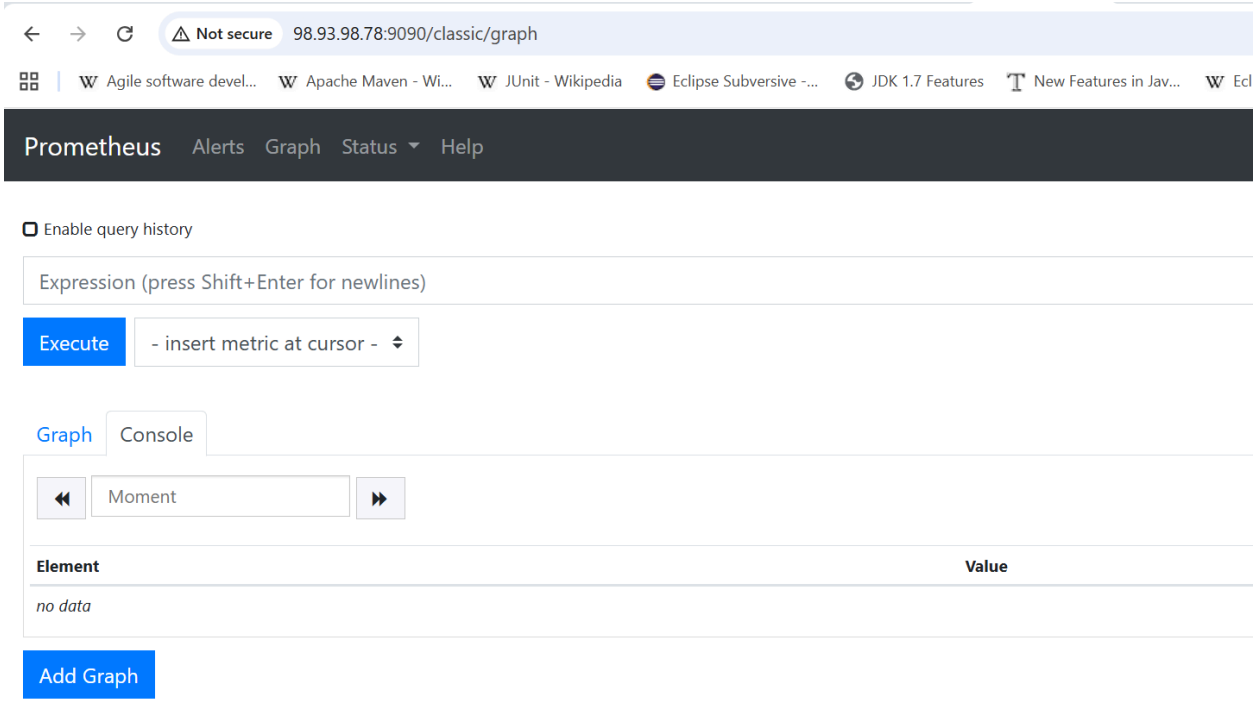
- SSH → 22
- HTTP → 9090 (Prometheus)
- Custom TCP → 3000 (Grafana)
- Custom TCP → 9100 (Node Exporter)





Install Prometheus:

Sudo apt install prometheus



Install node exporter:

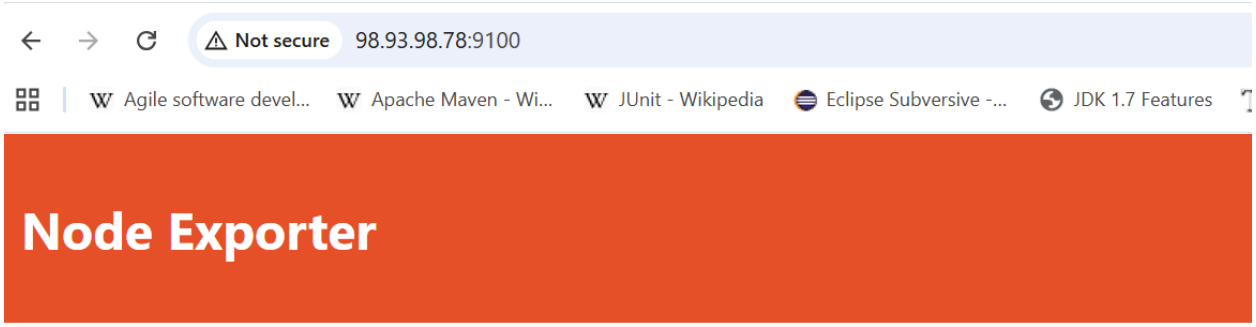
cd /tmp

wget https://github.com/prometheus/node_exporter/releases/download/v1.10.2/node_exporter-1.10.2.linux-amd64.tar.gz

tar -xvf node_exporter-1.10.2.linux-amd64.tar.gz

cd node_exporter-1.10.2.linux-amd64

./node_exporter



Prometheus Node Exporter

Version: (version=1.7.0, branch=debian/sid, revision=1.7.0-1ubuntu0.3)

- Metrics

sudo nano /etc/prometheus/prometheus.yml

```
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: 'prometheus'

    # Override the global default and scrape targets from this job every 5 seconds.
    scrape_interval: 5s
    scrape_timeout: 5s

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ['localhost:9090']

  - job_name: node
    # If prometheus-node-exporter is installed, grab stats about the local
    # machine by default.
    static_configs:
      - targets: ['localhost:9100']
```

Install grafana:

sudo apt update

sudo apt install grafana -y

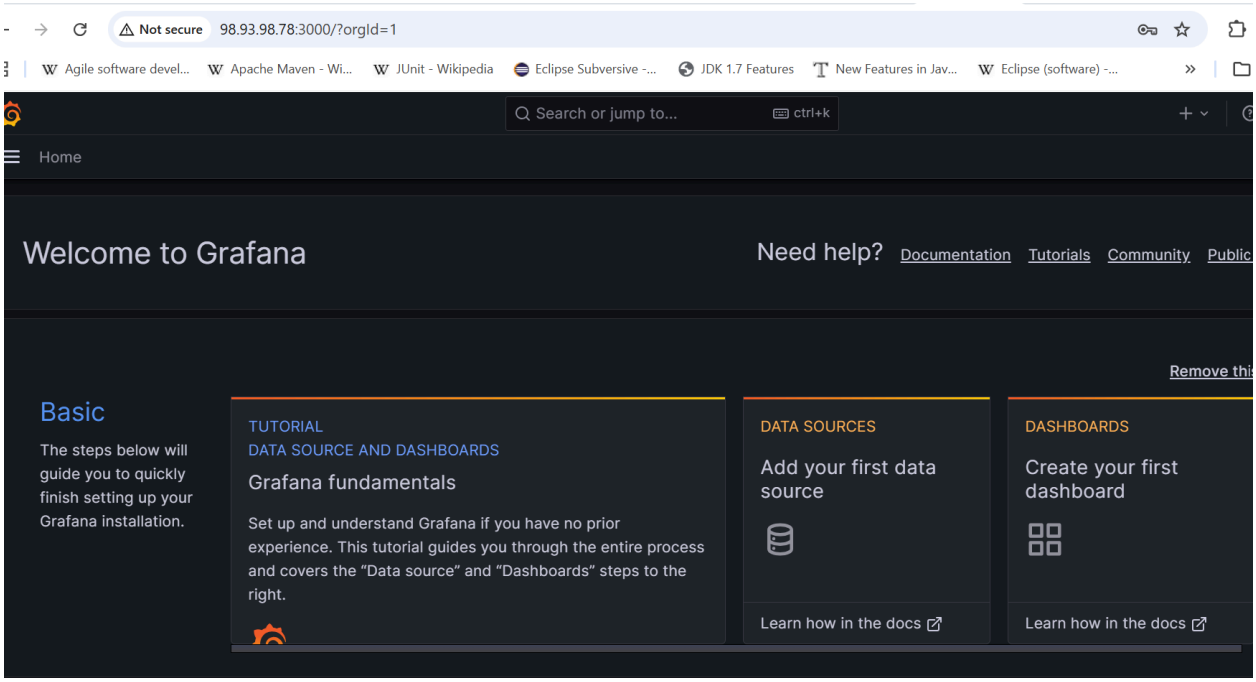
sudo systemctl start grafana-server

sudo systemctl enable grafana-server

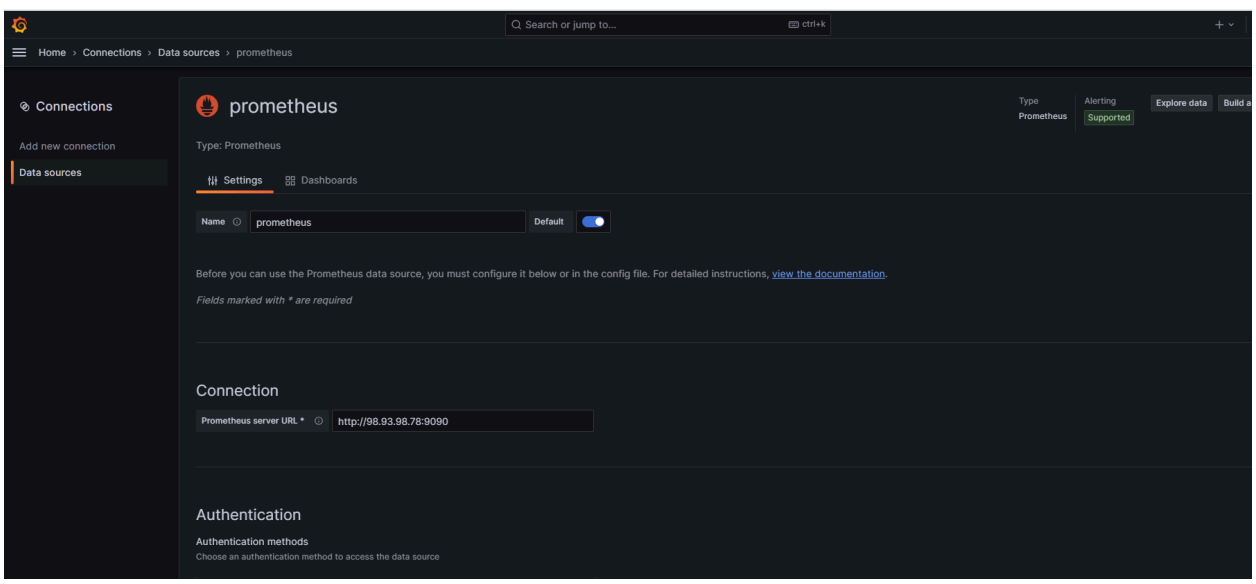
Login:

User:admin

Pwd:admin



Add data source as prometheus:



Create dashboard: dashboards – visualization – select datasource(prometheus) add query:

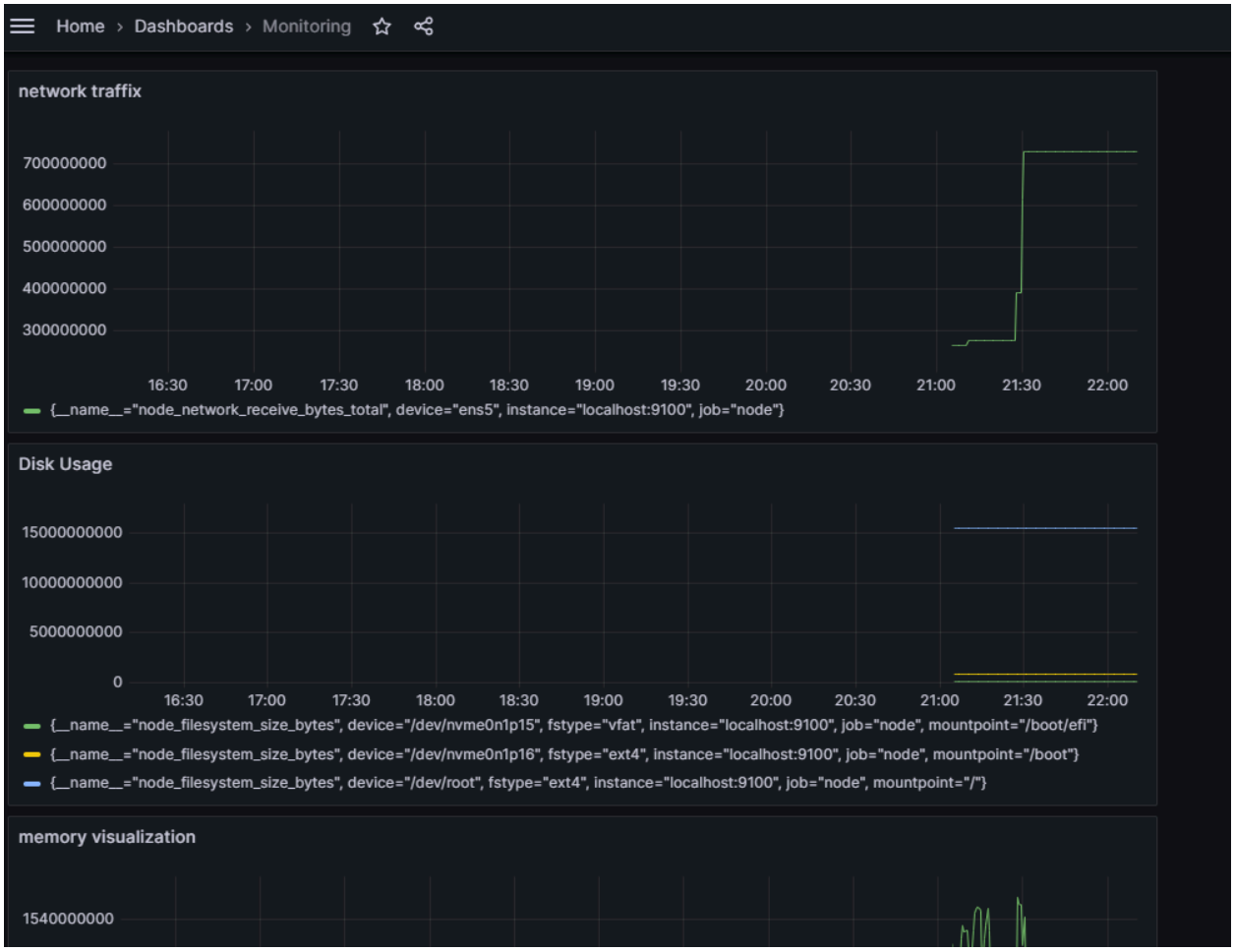
Queries

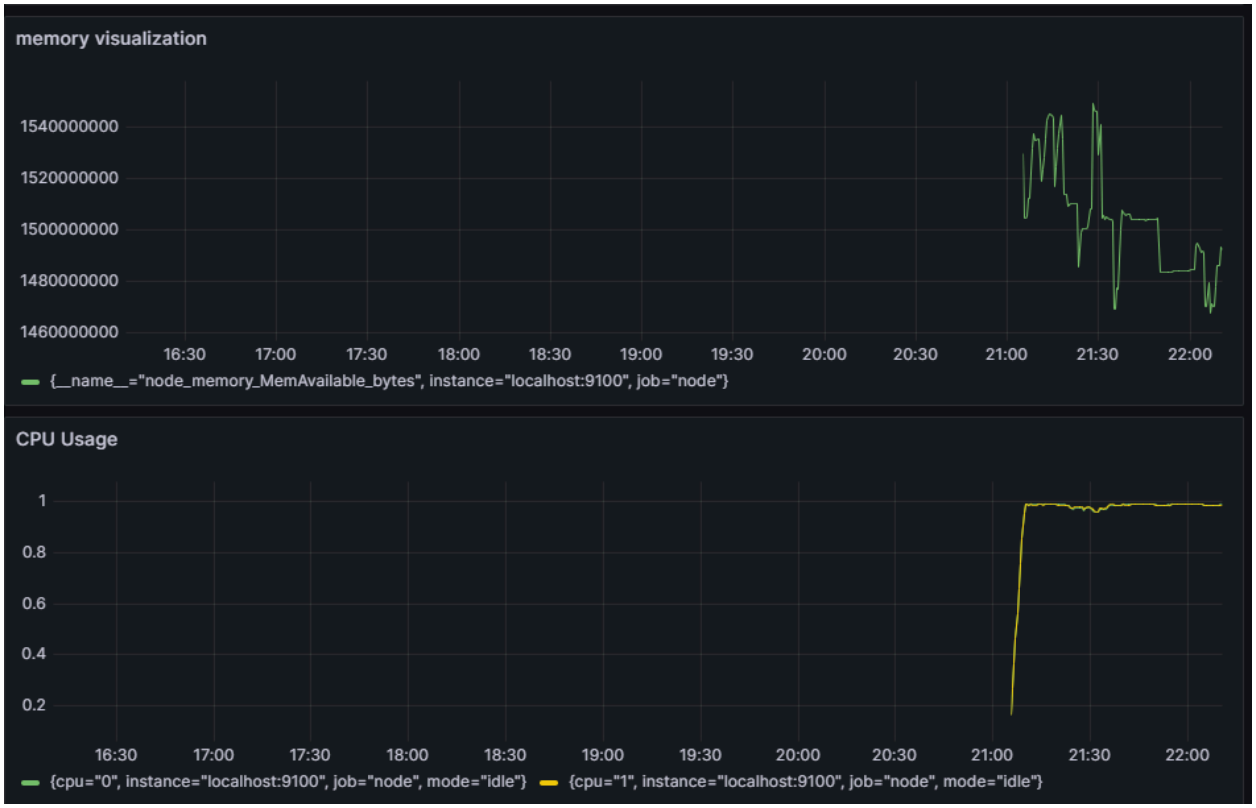
CPU Usage : rate(node_cpu_seconds_total{mode="idle"}[5m])

Memory Usage : node_memory_MemAvailable_bytes

Disk Usage : node_filesystem_size_bytes

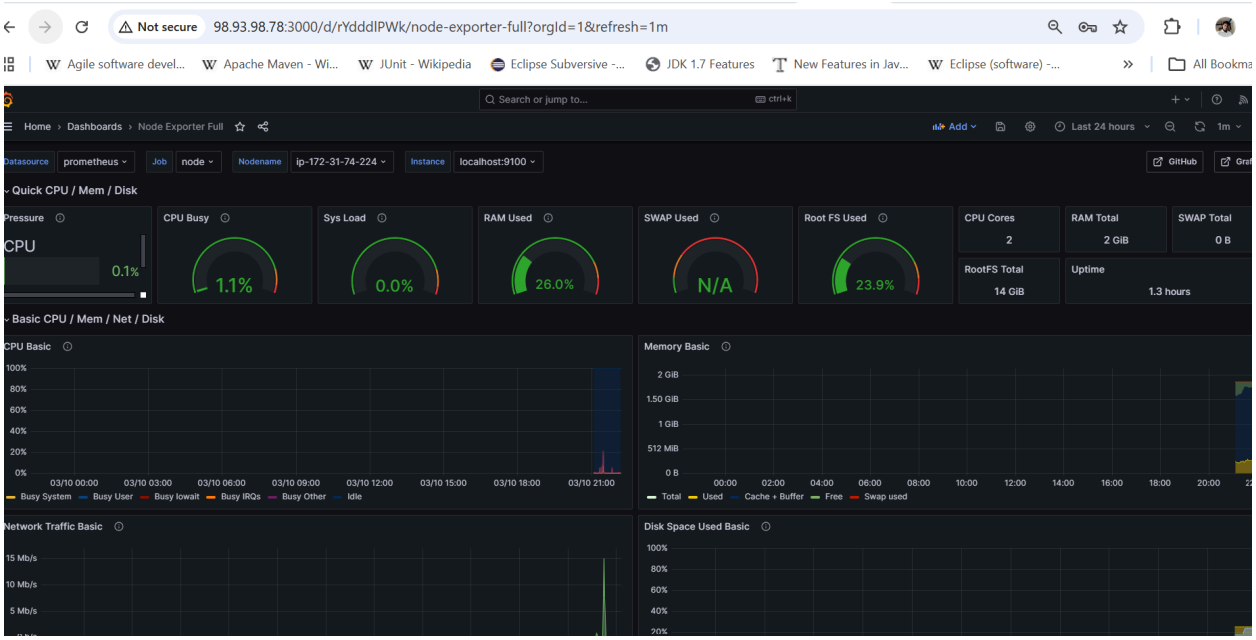
Network Traffic : node_network_receive_bytes_total





Grafana has ready dashboards for Node Exporter:

1. Go to **Dashboards → Import**
2. Enter dashboard ID: 1860


Not secure
98.93.98.78:3000/d/rYdddlPWk/node-exporter-full?orgId=1&refresh=1m
Agile software devel...
Apache Maven - Wi...
JUnit - Wikipedia
Eclipse Subversive -...
JDK 1.7 Features
New Features in Jav...
Eclipse (software) -...
All Bookma
Search or jump to...
ctrl+k
Home › Dashboards › Node Exporter Full
Add
Last 24 hours
1m
Datasource
prometheus
Job
node
Nodename
ip-172-31-74-224
Instance
localhost:9100
GitHub
Quick CPU / Mem / Disk
Pressure
CPU
0.1%
CPU Busy
1.1%
Sys Load
0.0%
RAM Used
26.0%
SWAP Used
N/A
Root FS Used
23.9%
CPU Cores
2
RAM Total
2 GiB
SWAP Total
0 B
RootFS Total
14 GiB
Uptime
1.3 hours
Basic CPU / Mem / Net / Disk
CPU Basic
100%
80%
60%
40%
20%
0%
03/10 00:00
03/10 03:00
03/10 06:00
03/10 09:00
03/10 12:00
03/10 15:00
03/10 18:00
03/10 21:00
Busy System
Busy User
Busy Iowait
Busy IRQs
Busy Other
Idle
Memory Basic
2 GiB
1.50 GiB
1 GiB
512 MiB
0 B
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
Total
Used
Cache + Buffer
Free
Swap used
Network Traffic Basic
15 Mb/s
10 Mb/s
5 Mb/s
Disk Space Used Basic
100%
80%
60%
40%